

K2-29b

R-2280

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-29_210107 · created 2026-08-17T00:02:16+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459221.7722 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.047
Transit depth (Rp/Rs)^2	2.96 +/- 1.63 [%]
Orbital Inclination (inc)	85.04 +/- 1.91
Airmass coefficient 1 (a1)	0.991 +/- 0.031
Airmass coefficient 2 (a2)	0.007 +/- 0.014
Scatter in the residuals of the lightcurve fit is	3.1 %
Best Comparison Star	#3 - [579, 163]
Optimal Aperture	4.78
Optimal Annulus	17.11
Transit Duration (day)	0.037 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.172 ± 0.047 vs 0.1419 (deviation 0.57σ)
Duration fit vs published	0.037 d vs 0.0925 d (deviation 2.46σ)
Mid-time O–C	-10.8 min
Depth SNR	3.2
Residual scatter	3.1 %

Light curve

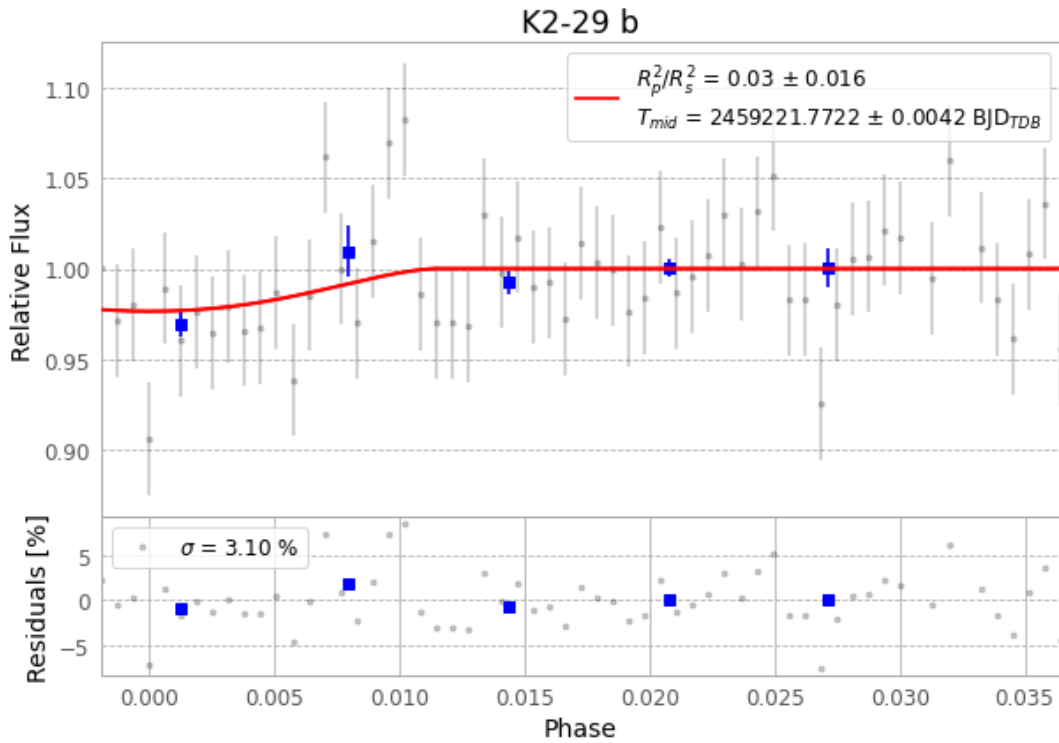


Figure 1 — FinalLightCurve_K2-29 b_2021-01-06.png (SHA-256 f09baec967aeb3f6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [370, 277], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 454ba0bd18d87da8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

61 FITS frames (40 MB), K2-29210107061512.FITS → K2-29210107091512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-29-210107.log (bd70095d90eb...), FinalLightCurve_K2-29 b_2021-01-06.png (f09baec967ae...), FinalLightCurve_K2-29 b_2021-01-06.csv (493dddf5311...)

Compute resources

Wall time 107.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 00:02Z.